Algebraic Geometry 15

Riemann surfaces

Contents

1 Problem 6: The Riemann Surface of $(z^4 - 1)^{1/4}$

Problem statement

Let
$$\pi : R \to \mathbb{C} \setminus \{1, i, -1, -i\}$$
be the Riemann surface associated to the complete analytic function
$$(z^4 - 1)^{1/4}.$$
Describe R explicitly by a gluing construction. Assuming the fact that R may be compactified to a compact Riemann surface $\overline{R}$ and π extended to an analytic map
$$\overline{\pi} : \overline{R} \to \mathbb{C}_\infty,$$
find the genus of $\overline{R}$.

Solution

We study the multivalued function
$$w = (z^4 - 1)^{1/4}.$$
Equivalently, its compactified Riemann surface is the smooth projective model of the affine curve
$$w^4 = z^4 - 1.$$
The branch points in the finite z-plane are the roots of
$$z^4 - 1 = 0,$$
namely
$$1, \quad i, \quad -1, \quad -i.$$
Away from these four points, the equation
$$w^4 = z^4 - 1$$
has four distinct values of w. Therefore over
$$\mathbb{C} \setminus \{1, i, -1, -i\}$$
the surface is a four-sheeted covering.

Choose four copies
$$S_0, S_1, S_2, S_3$$
of the cut plane
$$\mathbb{C} \setminus \Gamma,$$

where Γ is a system of cuts joining the branch points. For example, we may take cuts joining the points cyclically:

$$1 \longleftrightarrow i, \qquad i \longleftrightarrow -1, \qquad -1 \longleftrightarrow -i, \qquad -i \longleftrightarrow 1.$$

On the k-th sheet we choose the branch

$$w_k(z) = e^{2\pi ik/4}(z^4-1)^{1/4} = i^k(z^4-1)^{1/4}, \qquad k = 0, 1, 2, 3.$$

When one analytically continues once around any branch point, the value of $(z^4-1)^{1/4}$ is multiplied by

$$e^{2\pi i/4} = i.$$

Thus the monodromy sends

$$S_k \longmapsto S_{k+1},$$

with indices taken modulo 4.

Hence the gluing rule is:

$$\boxed{\text{the upper bank of each cut on } S_k \text{ is glued to the lower bank of the same cut on } S_{k+1}.}$$

Equivalently, the four sheets are glued cyclically:

$$S_0 \to S_1 \to S_2 \to S_3 \to S_0.$$

Near a branch point $a \in \{1, i, -1, -i\}$, we have

$$z^4 - 1 = (z-a)u(z),$$

where $u(a) \neq 0$. Therefore locally

$$w^4 = (z-a)u(z).$$

After absorbing the unit $u(z)$, this has the local form

$$z - a = t^4.$$

So over each branch point there is one ramification point with ramification index

$$e = 4.$$

There are four such branch points. Therefore the total ramification contribution is

$$4(e-1) = 4(4-1) = 12.$$

The map

$$\overline{\pi} : \overline{R} \to \mathbb{C}_\infty$$

has degree 4. By the Riemann–Hurwitz formula,

$$2g(\overline{R}) - 2 = 4(2g(\mathbb{C}_\infty) - 2) + \sum_P (e_P - 1).$$

Since

$$g(\mathbb{C}_\infty) = 0,$$

we get

$$2g(\overline{R}) - 2 = 4(-2) + 12.$$

Thus

$$2g(\overline{R}) - 2 = 4,$$

so

$$2g(\overline{R}) = 6,$$

and therefore

$$\boxed{g(\overline{R}) = 3.}$$

Thus the compactified Riemann surface of

$$(z^4 - 1)^{1/4}$$

has genus 3.

2 Problem 7: Classification of Discrete Additive Subgroups of $\mathbb{C}$

Problem statement

Suppose $\Omega \subseteq \mathbb{C}$ is a discrete additive subgroup. Here, discrete means that the subspace topology on Ω is discrete. Show that one of the following holds:

$$\text{(i)} \quad \Omega = \{0\},$$

or

$$\text{(ii)} \quad \Omega = \mathbb{Z}\omega \quad \text{for some } \omega \neq 0,$$

or

$$\text{(iii)} \quad \Omega = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$$

with

$$\omega_1, \omega_2 \neq 0$$

and

$$\omega_2/\omega_1 \notin \mathbb{R}.$$

Solution

We identify

$$\mathbb{C} \cong \mathbb{R}^2.$$

Thus Ω is a discrete additive subgroup of the real vector space $\mathbb{R}^2$.

If

$$\Omega = \{0\},$$

we are in case (i).

Assume now that

$$\Omega \neq \{0\}.$$

Since Ω is discrete, there exists a nonzero element of minimal length. Choose

$$\omega_1 \in \Omega, \qquad \omega_1 \neq 0,$$

such that

$$|\omega_1| \leq |\omega|$$

for every nonzero $\omega \in \Omega$.

First suppose that every element of Ω lies on the real line spanned by ω_1. Then

$$\Omega \subseteq \mathbb{R}\omega_1.$$

We claim that

$$\Omega = \mathbb{Z}\omega_1.$$

Indeed, take any $\omega \in \Omega$. Since $\omega \in \mathbb{R}\omega_1$, there exists $t \in \mathbb{R}$ such that

$$\omega = t\omega_1.$$

Choose $n \in \mathbb{Z}$ such that

$$t - n \in \left[-\frac{1}{2}, \frac{1}{2}\right].$$

Then

$$\omega - n\omega_1 = (t-n)\omega_1 \in \Omega.$$

But

$$|\omega - n\omega_1| \leq \frac{1}{2}|\omega_1|.$$

By minimality of ω_1, this is possible only if

$$\omega - n\omega_1 = 0.$$

Therefore

$$\omega = n\omega_1.$$

Hence

$$\Omega = \mathbb{Z}\omega_1,$$

which is case (ii).

Now suppose that Ω contains an element not lying on

$$\mathbb{R}\omega_1.$$

Then choose

$$\omega_2 \in \Omega \setminus \mathbb{R}\omega_1$$

with minimal positive distance from the line $\mathbb{R}\omega_1$. This is possible because Ω is discrete.

We claim that

$$\Omega = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2.$$

Let $\omega \in \Omega$. Since ω_1, ω_2 are linearly independent over $\mathbb{R}$, there exist real numbers a, b such that

$$\omega = a\omega_1 + b\omega_2.$$

Choose integers $m, n \in \mathbb{Z}$ such that

$$a - m \in \left[-\frac{1}{2}, \frac{1}{2}\right], \qquad b - n \in \left[-\frac{1}{2}, \frac{1}{2}\right].$$

Then

$$\eta = \omega - m\omega_1 - n\omega_2 \in \Omega.$$

Moreover, η lies in a bounded fundamental parallelogram spanned by ω_1, ω_2.

If $\eta \neq 0$, then by subtracting suitable integer multiples of ω_1, one would obtain a nonzero element of Ω either shorter than ω_1 in the direction of ω_1, or with smaller positive distance from the line $\mathbb{R}\omega_1$ than ω_2. This contradicts the choices of ω_1 and ω_2.

Hence

$$\eta = 0.$$

Therefore

$$\omega = m\omega_1 + n\omega_2.$$

So

$$\Omega = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2.$$

Because $\omega_2 \notin \mathbb{R}\omega_1$, we have

$$\omega_2/\omega_1 \notin \mathbb{R}.$$

Therefore Ω is either trivial, cyclic, or a rank-two lattice:

$$\boxed{\Omega = \{0\}, \quad \Omega = \mathbb{Z}\omega, \quad \text{or} \quad \Omega = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 \text{ with } \omega_2/\omega_1 \notin \mathbb{R}.}$$

3 Problem 8: Finite Fourier Expansion of a Simply Periodic Analytic Function

Problem statement

Let f be a simply periodic analytic function on $\mathbb{C}$ with periods $\mathbb{Z}$. Suppose furthermore that

$$f(x + iy)$$

converges uniformly in x to possibly infinite limits as

$$y \to \pm\infty.$$

Show that

$$f(z) = \sum_{k=-n}^{m} a_k e^{2\pi i k z},$$

i.e. $f(z)$ has a finite Fourier expansion.

Solution

Since f has periods $\mathbb{Z}$, we have

$$f(z+1) = f(z).$$

Therefore f descends to an analytic function on the punctured plane

$$\mathbb{C}^*.$$

Define

$$q = e^{2\pi i z}.$$

If

$$z = x + iy,$$

then

$$|q| = |e^{2\pi i(x+iy)}| = e^{-2\pi y}.$$

Hence

$$y \to +\infty \quad \iff \quad q \to 0,$$

and

$$y \to -\infty \quad \iff \quad q \to \infty.$$

Because f is 1-periodic, there exists an analytic function

$$F : \mathbb{C}^* \to \mathbb{C}$$

such that

$$f(z) = F(q) = F(e^{2\pi i z}).$$

The assumption that $f(x+iy)$ converges uniformly in x as $y \to +\infty$ says that $F(q)$ has a limit as

$$q \to 0.$$

This limit is allowed to be infinite. Therefore $q = 0$ is either a removable singularity or a pole of F.

Similarly, the assumption that $f(x+iy)$ converges uniformly in x as $y \to -\infty$ says that $F(q)$ has a limit as

$$q \to \infty.$$

Thus $q = \infty$ is either a removable singularity or a pole of F.

Therefore F is meromorphic on the Riemann sphere

$$\mathbb{C}_\infty$$

with possible poles only at

$$0 \quad \text{and} \quad \infty.$$

A meromorphic function on $\mathbb{C}_\infty$ with poles only at 0 and ∞ is a Laurent polynomial. Hence there exist integers $m, n \geq 0$ and constants $a_k \in \mathbb{C}$ such that

$$F(q) = \sum_{k=-n}^{m} a_k q^k.$$

Substituting

$$q = e^{2\pi i z},$$

we obtain

$$f(z) = F(e^{2\pi i z}) = \sum_{k=-n}^{m} a_k e^{2\pi i k z}.$$

Thus f has a finite Fourier expansion:

$$\boxed{f(z) = \sum_{k=-n}^{m} a_k e^{2\pi i k z}.}$$

4 Problem 9: Zeros and Poles of an Elliptic Function

Problem statement

Let f be a non-constant elliptic function with respect to a lattice

$$\Lambda \subset \mathbb{C},$$

and let $P \subset \mathbb{C}$ be a fundamental parallelogram. Using the argument principle and, if necessary, slightly perturbing P, show that the number of zeros of f in P is the same as the number of poles, both counted with multiplicities.

This is also a consequence of the valency theorem, but the point of this question is that this more direct argument via contour integration also works.

Solution

Let

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2,$$

where

$$\omega_2/\omega_1 \notin \mathbb{R}.$$

A fundamental parallelogram is

$$P = \{a + s\omega_1 + t\omega_2 : 0 \leq s, t \leq 1\}.$$

Because f is elliptic with respect to Λ, we have

$$f(z + \omega) = f(z)$$

for every

$$\omega \in \Lambda.$$

In particular,

$$f(z + \omega_1) = f(z), \qquad f(z + \omega_2) = f(z).$$

If necessary, slightly translate the parallelogram P so that no zero or pole of f lies on its boundary. This is possible because zeros and poles of a meromorphic function are isolated.

By the argument principle,

$$N - P = \frac{1}{2\pi i} \int_{\partial P} \frac{f'(z)}{f(z)} \, dz,$$

where N is the number of zeros of f inside P, counted with multiplicity, and P is the number of poles of f inside P, counted with multiplicity.

We now show that the boundary integral is zero.

The boundary of the parallelogram consists of four sides. Opposite sides are translations of each other by periods. Since

$$f(z + \omega_j) = f(z),$$

we also have

$$f'(z + \omega_j) = f'(z).$$

Therefore

$$\frac{f'(z + \omega_j)}{f(z + \omega_j)} = \frac{f'(z)}{f(z)}.$$

Thus the logarithmic derivative

$$\frac{f'}{f}$$

is also elliptic with respect to the same lattice.

Now consider the two opposite sides parallel to ω_1. They are translated versions of one another by ω_2, but they are traversed in opposite directions in the boundary orientation. Since $\frac{f'}{f}$ is periodic, their integrals cancel.

Similarly, the two opposite sides parallel to ω_2 are translated versions of one another by ω_1, again traversed in opposite directions. Their integrals also cancel.

Therefore

$$\int_{\partial P} \frac{f'(z)}{f(z)} \, dz = 0.$$

Hence the argument principle gives

$$N - P = 0.$$

Therefore

$$N = P.$$

Thus the number of zeros of f in a fundamental parallelogram equals the number of poles, counted with multiplicities:

$$\boxed{\#\{\text{zeros of } f \text{ in } P\} = \#\{\text{poles of } f \text{ in } P\}.}$$

5 Theory Behind Problems 6–9: Branched Coverings, Lattices, Fourier Expansions, and Elliptic Functions

Problems 6–9 form a natural progression through several of the central ideas of complex analysis and Riemann surface theory:

Riemann Surfaces $\longrightarrow$ Branched Coverings $\longrightarrow$ Lattices $\longrightarrow$ Complex Tori $\longrightarrow$ Periodic Functions $\longrightarrow$

These concepts eventually lead to elliptic curves, algebraic geometry, modular forms, and number theory.

5.1 Branched Coverings and Multivalued Functions

Many complex functions are naturally multivalued. Examples include

$$\sqrt{z}, \qquad \log z, \qquad (z^4-1)^{1/4}.$$

For instance,

$$w = (z^4-1)^{1/4}$$

satisfies

$$w^4 = z^4 - 1.$$

At a generic point z, there are four possible values of w:

$$w, \quad iw, \quad -w, \quad -iw.$$

To make the function single-valued we construct a Riemann surface consisting of four sheets glued together appropriately.

5.2 Branch Points

A branch point occurs where several sheets meet.

For

$$w^4 = z^4 - 1,$$

branching occurs where

$$z^4 - 1 = 0.$$

Thus the branch points are

$$1, \qquad i, \qquad -1, \qquad -i.$$

Near such a point a,

$$w^4 = z - a.$$

Equivalently,

$$z = a + w^4.$$

The local model is therefore

$$w \longmapsto w^4.$$

Four sheets merge together at each branch point.

5.3 Monodromy

Monodromy describes what happens when one analytically continues a branch around a loop.

If we travel once around a branch point of

$$(z^4 - 1)^{1/4},$$

the value changes by

$$w \mapsto iw.$$

A second turn gives

$$w \mapsto -w,$$

a third turn gives

$$w \mapsto -iw,$$

and a fourth turn returns to the original branch.

Thus the monodromy permutation is

$$(1\,2\,3\,4).$$

Monodromy records how the sheets of a covering are glued together.

5.4 Ramification

The ramification index measures how many sheets come together.

If locally

$$z = t^e,$$

then the ramification index is e.

Examples:

$$z = t^2 \qquad \Rightarrow \qquad e = 2,$$

$$z = t^3 \qquad \Rightarrow \qquad e = 3,$$

$$z = t^4 \qquad \Rightarrow \qquad e = 4.$$

For Problem 6 each branch point has

$$e = 4.$$

5.5 The Riemann–Hurwitz Formula

The genus of a branched covering is determined by the Riemann–Hurwitz formula:

$$2g(X) - 2 = d\Big(2g(Y) - 2\Big) + \sum_P (e_P - 1),$$

where

- d is the degree of the covering,
- $g(X)$ is the genus upstairs,
- $g(Y)$ is the genus downstairs,
- e_P is the ramification index.

This formula shows precisely how branching creates additional handles on a surface.

For Problem 6:

$$d = 4, \qquad g(\mathbb{P}^1) = 0, \qquad \sum (e_P - 1) = 4(4-1) = 12.$$

Hence

$$2g - 2 = 4(-2) + 12 = 4,$$

which gives

$$g = 3.$$

Thus the compactified surface has genus three.

5.6 Discrete Additive Subgroups of $\mathbb{C}$

Problem 7 classifies discrete additive subgroups of $\mathbb{C}$.

A subgroup

$$\Omega \subseteq \mathbb{C}$$

is discrete if every point is isolated.

The classification theorem states:

$$\Omega = \{0\},$$

or

$$\Omega = \mathbb{Z}\omega,$$

or

$$\Omega = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2, \qquad \omega_2/\omega_1 \notin \mathbb{R}.$$

No other possibilities exist.

Since

$$\mathbb{C} \cong \mathbb{R}^2,$$

the maximum rank of a discrete subgroup is 2.

5.7 Lattices

A lattice is a rank-two discrete subgroup

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2,$$

where

$$\omega_2/\omega_1 \notin \mathbb{R}.$$

Geometrically, lattice points form a repeating grid in the complex plane.

Examples:

$$\Lambda = \mathbb{Z} + i\mathbb{Z},$$

(square lattice),

and

$$\Lambda = \mathbb{Z} + e^{i\pi/3}\mathbb{Z},$$

(hexagonal lattice).

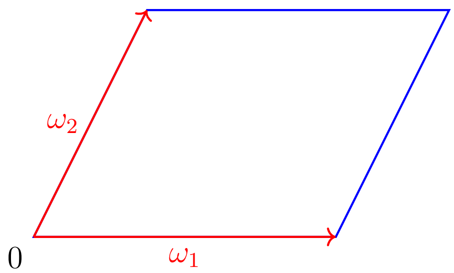
5.8 Fundamental Parallelogram

The fundamental parallelogram of a lattice is

$$P = \{s\omega_1 + t\omega_2 : 0 \leq s, t \leq 1\}.$$

Every point of $\mathbb{C}$ is congruent modulo Λ to exactly one point of P.

The complex plane is tiled by translated copies of P.



5.9 Complex Tori

By identifying opposite sides of a fundamental parallelogram,

$$\mathbb{C}/\Lambda,$$

we obtain a compact Riemann surface.

First glue left and right edges to form a cylinder.

Then glue top and bottom edges.

The result is a torus.

$$\boxed{\mathbb{C}/\Lambda \cong \text{torus.}}$$

This torus has genus

$$g = 1.$$

5.10 Periodic Functions

A function is periodic if

$$f(z+\omega) = f(z).$$

If only one period exists, for example

$$f(z+1) = f(z),$$

the function is called simply periodic.

If two independent periods exist,

$$f(z+\omega_1) = f(z), \qquad f(z+\omega_2) = f(z),$$

the function is doubly periodic.

5.11 Fourier Expansions

For a 1-periodic analytic function,

$$f(z+1) = f(z),$$

we introduce

$$q = e^{2\pi i z}.$$

Then

$$z \sim z+1$$

corresponds to the same value of q.

Every periodic analytic function has a Fourier expansion

$$f(z) = \sum_{k=-\infty}^{\infty} a_k e^{2\pi i k z}.$$

Equivalently,

$$f(z) = \sum_{k=-\infty}^{\infty} a_k q^k.$$

5.12 Finite Fourier Expansions

Problem 8 assumes controlled behaviour as

$$y \to \pm\infty.$$

Since

$$q = e^{2\pi i z},$$

we have

$$y \to +\infty \Longleftrightarrow q \to 0,$$

and

$$y \to -\infty \Longleftrightarrow q \to \infty.$$

The assumptions imply that $q = 0$ and $q = \infty$ are at worst poles.
Hence only finitely many positive and negative powers occur.
Therefore

$$f(z) = \sum_{k=-n}^{m} a_k e^{2\pi i k z}.$$

Thus f is represented by a finite Laurent polynomial in q.

5.13 Elliptic Functions

An elliptic function is a meromorphic function satisfying

$$f(z+\omega_1) = f(z), \qquad f(z+\omega_2) = f(z),$$

for a lattice

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2.$$

Hence elliptic functions are precisely meromorphic functions on

$$\mathbb{C}/\Lambda.$$

Since

$$\mathbb{C}/\Lambda$$

is a torus, we obtain:

$$\boxed{\text{Elliptic functions} = \text{meromorphic functions on a torus.}}$$

This is one of the most important ideas in complex analysis.

5.14 The Argument Principle

If f is meromorphic inside a contour γ, then

$$\frac{1}{2\pi i}\int_\gamma \frac{f'(z)}{f(z)}\,dz = N - P,$$

where

- N is the number of zeros,
- P is the number of poles,

counted with multiplicities.

The logarithmic derivative

$$\frac{f'}{f}$$

records the net number of zeros minus poles.

5.15 Zeros and Poles of Elliptic Functions

Let P be a fundamental parallelogram.

Because

$$f(z+\omega_j) = f(z),$$

the logarithmic derivative is also periodic:

$$\frac{f'(z+\omega_j)}{f(z+\omega_j)} = \frac{f'(z)}{f(z)}.$$

Opposite sides of P contribute equal integrals with opposite orientations.

Therefore

$$\int_{\partial P} \frac{f'}{f}\,dz = 0.$$

Applying the argument principle gives

$$N - P = 0.$$

Hence

$$\boxed{N = P.}$$

An elliptic function has exactly as many zeros as poles in a fundamental parallelogram, counted with multiplicities.

5.16 The Valency Theorem

A stronger result states

$$\sum_{a \in P} \mathrm{ord}_a(f) = 0.$$

Positive orders correspond to zeros.

Negative orders correspond to poles.

Thus the total multiplicity of zeros equals the total multiplicity of poles.

Problem 9 is essentially a direct proof of this theorem using contour integration.

5.17 Conceptual Summary

$$\text{Branch Points} \longrightarrow \text{Ramification} \longrightarrow \text{Riemann Surfaces} \longrightarrow \text{Riemann–Hurwitz}$$

$$\text{Discrete Subgroups} \longrightarrow \text{Lattices} \longrightarrow \text{Fundamental Parallelograms} \longrightarrow \text{Complex Tori}$$

$$\text{Periodic Functions} \longrightarrow \text{Fourier Series} \longrightarrow \text{Elliptic Functions}$$

$$\text{Elliptic Functions} \longrightarrow \text{Argument Principle} \longrightarrow \text{Valency Theorem} \longrightarrow \text{Elliptic Curves.}$$

These ideas form one of the principal bridges between complex analysis, topology, Riemann surfaces, and modern algebraic geometry.

6 Introduction to Modular Forms

The theory of modular forms is one of the deepest continuations of the theory of lattices, complex tori, and elliptic functions.

The chain of ideas is

$$\text{Lattices} \longrightarrow \text{Complex Tori} \longrightarrow \text{Elliptic Functions} \longrightarrow \text{Modular Forms} \longrightarrow \text{Number Theory.}$$

Historically, modular forms arose from the study of elliptic integrals and elliptic functions, but today they play central roles in algebraic geometry, arithmetic geometry, representation theory, and mathematical physics.

6.1 The Upper Half-Plane

The fundamental object is the upper half-plane

$$\mathbb{H} = \{\tau \in \mathbb{C} : \mathrm{Im}(\tau) > 0\}.$$

Every lattice

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$$

may be rescaled so that

$$\omega_1 = 1, \qquad \omega_2 = \tau,$$

with

$$\tau \in \mathbb{H}.$$

Thus every lattice can be written in the form

$$\Lambda_\tau = \mathbb{Z} + \tau\mathbb{Z}.$$

Consequently, points of $\mathbb{H}$ parameterize complex tori

$$\mathbb{C}/\Lambda_\tau.$$

6.2 Equivalent Lattices

Different values of τ may determine isomorphic tori.

For example,

$$\Lambda_\tau = \mathbb{Z} + \tau\mathbb{Z}$$

and

$$\Lambda_{\tau+1} = \mathbb{Z} + (\tau+1)\mathbb{Z}$$

define the same lattice.

Likewise,

$$\Lambda_\tau \quad \text{and} \quad \Lambda_{-1/\tau}$$

define equivalent complex tori.

These transformations generate the modular group

$$SL_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{Z},\ ad - bc = 1 \right\}.$$

Its action on $\mathbb{H}$ is

$$\tau \mapsto \frac{a\tau + b}{c\tau + d}.$$

6.3 The Modular Group

The modular group acts by biholomorphic transformations of the upper half-plane.

The two fundamental generators are

$$T(\tau) = \tau + 1,$$

and

$$S(\tau) = -\frac{1}{\tau}.$$

Every element of $SL_2(\mathbb{Z})$ can be obtained from repeated compositions of S and T.

The quotient space

$$SL_2(\mathbb{Z}) \backslash \mathbb{H}$$

classifies complex tori up to conformal equivalence.

Thus the moduli space of elliptic curves is essentially

$$SL_2(\mathbb{Z}) \backslash \mathbb{H}.$$

6.4 Definition of a Modular Form

A modular form of weight k is a holomorphic function

$$f : \mathbb{H} \to \mathbb{C}$$

satisfying

$$f\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^k f(\tau)$$

for every

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}),$$

and satisfying an additional regularity condition at infinity.

The factor

$$(c\tau + d)^k$$

is called the automorphy factor.

6.5 The Cusp at Infinity

The point

$$\tau = i\infty$$

is called the cusp at infinity.

Because modular forms satisfy

$$f(\tau + 1) = f(\tau),$$

they are periodic.

We therefore introduce

$$q = e^{2\pi i\tau}.$$

This converts periodicity into ordinary power series.

A modular form has a Fourier expansion

$$f(\tau) = \sum_{n=0}^{\infty} a_n q^n.$$

This is called the q-expansion.

The coefficients a_n often contain deep arithmetic information.

6.6 Cusp Forms

If

$$a_0 = 0,$$

then f is called a cusp form.

Equivalently,

$$f(i\infty) = 0.$$

Cusp forms vanish at the cusp and form one of the most important spaces in number theory.

6.7 Eisenstein Series

The simplest modular forms are Eisenstein series.

For even integers $k \geq 4$,

$$G_k(\tau) = \sum_{(m,n)\neq(0,0)} \frac{1}{(m+n\tau)^k}.$$

These series converge absolutely and define modular forms of weight k.

Examples:

$$G_4(\tau), \qquad G_6(\tau).$$

Every classical modular form can ultimately be built from these.

6.8 The Modular Discriminant

One of the most important cusp forms is

$$\Delta(\tau) = q \prod_{n=1}^{\infty} (1-q^n)^{24}.$$

Its Fourier expansion begins

$$\Delta(\tau) = q - 24q^2 + 252q^3 - 1472q^4 + \cdots .$$

This is a cusp form of weight 12.

The coefficients of Δ contain profound arithmetic information.

6.9 The Modular j-Invariant

A central object is the modular invariant

$$j(\tau).$$

It is constructed from Eisenstein series and the discriminant:

$$j(\tau) = 1728 \frac{G_4(\tau)^3}{G_4(\tau)^3 - 27G_6(\tau)^2}.$$

The remarkable theorem is:

$$j(\tau_1) = j(\tau_2)$$

if and only if

$$\Lambda_{\tau_1}$$

and

$$\Lambda_{\tau_2}$$

define isomorphic complex tori.

Thus j completely classifies elliptic curves over $\mathbb{C}$.

6.10 Connection with Elliptic Curves

Every lattice

$$\Lambda$$

determines an elliptic curve

$$E_\Lambda = \mathbb{C}/\Lambda.$$

Using the Weierstrass $\wp$-function,

$$\wp(z),$$

one obtains an algebraic equation

$$y^2 = 4x^3 - g_2 x - g_3.$$

The coefficients

$$g_2, \qquad g_3$$

are modular forms.

Thus modular forms naturally describe families of elliptic curves.

6.11 Modular Forms and Number Theory

One reason modular forms are so important is that their Fourier coefficients encode arithmetic information.

Examples include:

- partitions,
- divisor sums,
- representations by quadratic forms,
- class numbers,
- values of L-functions,
- arithmetic of elliptic curves.

Many famous number-theoretic identities arise from modular forms.

6.12 The Modularity Theorem

A milestone of modern mathematics is the Modularity Theorem.

It states that every elliptic curve over $\mathbb{Q}$ is associated with a modular form.

Symbolically,

$$\text{Elliptic Curve} \longleftrightarrow \text{Modular Form}.$$

This theorem was the key ingredient in the proof of Fermat's Last Theorem.

6.13 Geometric Picture

The progression of ideas can be visualized as

$$\tau \in \mathbb{H}$$

$$\Downarrow$$

$$\Lambda_\tau = \mathbb{Z} + \tau\mathbb{Z}$$

$$\Downarrow$$

$$\mathbb{C}/\Lambda_\tau$$

$$\Downarrow$$

$$\text{Complex Torus}$$

$$\Downarrow$$

$$\text{Elliptic Curve}$$

$$\Downarrow$$

$$\text{Modular Forms}$$

$$\Downarrow$$

$$\text{Arithmetic Information.}$$

6.14 Conceptual Summary

$$\text{Discrete Subgroups} \longrightarrow \text{Lattices}$$

$$\text{Lattices} \longrightarrow \text{Complex Tori}$$

$$\text{Complex Tori} \longrightarrow \text{Elliptic Functions}$$

$$\text{Elliptic Functions} \longrightarrow \text{Elliptic Curves}$$

$$\text{Elliptic Curves} \longrightarrow \text{Modular Forms}$$

$$\text{Modular Forms} \longrightarrow \text{Number Theory.}$$

Modular forms may be viewed as functions on the space of all complex tori. They encode how elliptic curves vary, and their Fourier coefficients reveal some of the deepest arithmetic structures known in mathematics.

7 Visual Diagrams, Tables, and Concept Maps for Problems 6–9

7.1 Roadmap of the Theory

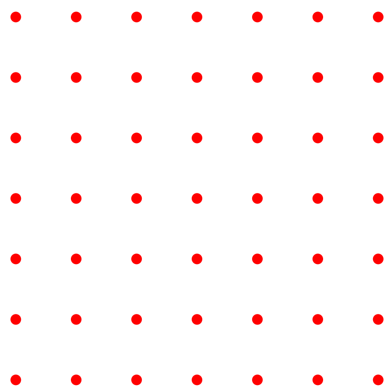
$$
\begin{tikzcd}
\text{Branched Coverings} \arrow[d] \\
\text{Riemann Surfaces} \arrow[d] \\
\text{Lattices} \arrow[d] \\
\text{Complex Tori} \arrow[d] \\
\text{Elliptic Functions} \arrow[d] \\
\text{Modular Forms}
\end{tikzcd}
$$

7.2 Classification of Discrete Subgroups

Rank	Form	Geometry	Example
0	$\{0\}$	Point	$\{0\}$
1	$\mathbb{Z}\omega$	Line of points	$\mathbb{Z}$
2	$\mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$	Lattice	$\mathbb{Z} + i\mathbb{Z}$

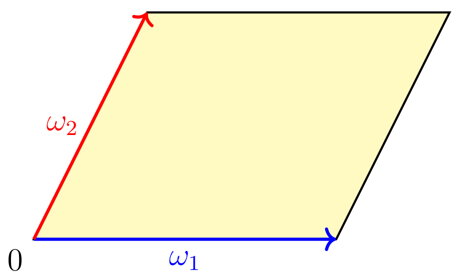
7.3 Rank 1 Versus Rank 2

Rank 1



Rank 2 Lattice

7.4 Fundamental Parallelogram



The shaded region is one copy of the torus before edge identifications.

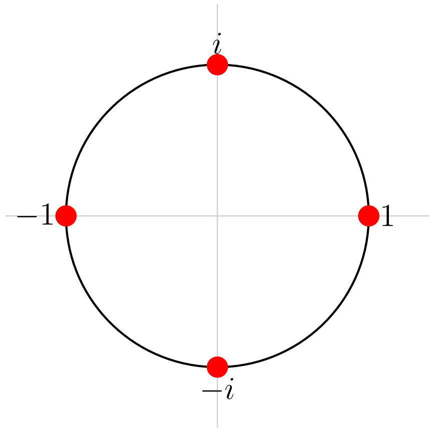
7.5 From Parallelogram to Torus

$$\boxed{\text{Parallelogram}} \xrightarrow{\text{Glue Left/Right}} \boxed{\text{Cylinder}} \xrightarrow{\text{Glue Top/Bottom}} \boxed{\text{Torus}}$$

7.6 Branching in Problem 6

Branch points:

$$1, \quad i, \quad -1, \quad -i.$$



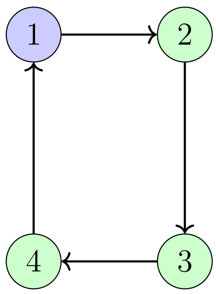
Each red point is a branch point where four sheets meet.

7.7 Monodromy of Problem 6

$$w = (z^4 - 1)^{1/4}$$

A loop around any branch point produces

$$w \to iw \to -w \to -iw \to w.$$



Monodromy permutation:

$$(1\,2\,3\,4).$$

7.8 Genus Comparison

Surface	Genus	Shape
Sphere	0	No holes
Torus	1	One hole
Double Torus	2	Two holes
Problem 6 Surface	3	Three holes

7.9 Periodic and Elliptic Functions

7.10 Fourier Expansion

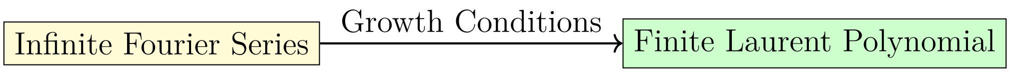
For a periodic function:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n e^{2\pi i n z}.$$

Problem 8 shows:

$$f(z) = \sum_{n=-N}^{M} a_n e^{2\pi i n z}.$$

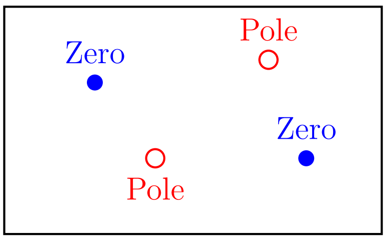
only finitely many terms survive.



7.11 Zeros and Poles in a Fundamental Parallelogram

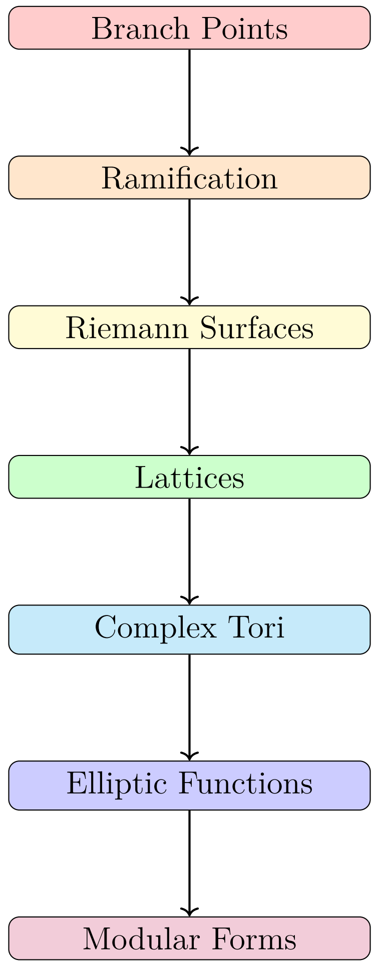
Problem 9 proves

$$N = P.$$



The total multiplicity of zeros equals the total multiplicity of poles.

7.12 Complete Concept Map



This diagram summarizes the entire mathematical journey from Problem 6 through Problem 9 and onward to modular forms.

8 Sheaves, Presheaves, Sheafification, and Stalks

8.1 Motivation

Many geometric and analytic objects are naturally defined only on open subsets of a space:

- continuous functions,
- differentiable functions,
- holomorphic functions,
- vector fields,
- differential forms,
- solutions of differential equations.

A sheaf formalizes the principle that global information can be reconstructed from compatible local information.

$$
\begin{tikzcd}
\boxed{\text{Local Data on Open Sets}} \arrow[d] \\
\boxed{\text{Compatibility on Overlaps}} \arrow[d] \\
\boxed{\text{Global Object}}
\end{tikzcd}
$$

8.2 Presheaves

Let X be a topological space.

Definition 8.2.1 (Presheaf)**.** A *presheaf* $\mathcal{F}$ on X consists of

1. a set (or group, ring, module)
$$\mathcal{F}(U)$$
for every open subset $U \subseteq X$,
2. restriction maps
$$\rho_{UV} : \mathcal{F}(U) \to \mathcal{F}(V)$$
for every inclusion $V \subseteq U$.

These satisfy:

1. Identity:
$$\rho_{UU} = id.$$
2. Composition: if
$$W \subseteq V \subseteq U,$$
then
$$\rho_{UW} = \rho_{VW} \circ \rho_{UV}.$$

Usually we write

$$s|_V$$

instead of

$$\rho_{UV}(s).$$

8.3 Restriction Diagram

$$
\begin{tikzcd}
 & U \arrow[dl] \arrow[dd] \\
V \arrow[dr] & \\
 & W
\end{tikzcd}
$$

The direct restriction

$$U \to W$$

equals the composition

$$U \to V \to W.$$

8.4 Examples of Presheaves

Continuous Functions

$$\mathcal{C}(U) = \{f : U \to \mathbb{R} \mid f \text{ continuous}\}.$$

Restrictions are ordinary restrictions of functions.

Holomorphic Functions

For a complex manifold,

$$\mathcal{O}(U) = \{f : U \to \mathbb{C} \mid f \text{ holomorphic}\}.$$

Differential Forms

$$\Omega^k(U) = \{k\text{-forms on } U\}.$$

8.5 Why Presheaves Are Not Enough

A presheaf only provides restrictions.

It does not guarantee that local pieces can be glued together.

The missing ingredient leads to the notion of a sheaf.

8.6 Sheaves

Definition 8.6.1 (Sheaf)**.** A presheaf $\mathcal{F}$ is a *sheaf* if the following two axioms hold.

Locality

Suppose

$$U = \bigcup_i U_i.$$

If

$$s, t \in \mathcal{F}(U)$$

satisfy

$$s|_{U_i} = t|_{U_i}$$

for all i, then

$$s = t.$$

Gluing

Suppose

$$s_i \in \mathcal{F}(U_i)$$

satisfy

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$$

for all i, j.

Then there exists a unique

$$s \in \mathcal{F}(U)$$

such that

$$s|_{U_i} = s_i.$$

8.7 The Gluing Principle

$$\begin{tikzcd}
 & \boxed{\text{Global Section } s} \arrow[dl] \arrow[dr] & \\
\boxed{U_1} \arrow[dr] & & \boxed{U_2} \arrow[dl] \\
 & \boxed{U_1 \cap U_2} &
\end{tikzcd}$$

The local sections must agree on overlaps before a global section can exist.

8.8 Example: Continuous Functions Form a Sheaf

Let

$$U = U_1 \cup U_2.$$

Suppose

$$f_1 \in C(U_1), \qquad f_2 \in C(U_2),$$

and

$$f_1 = f_2$$

on

$$U_1 \cap U_2.$$

Then define

$$f(x) = \begin{cases} f_1(x), & x \in U_1, \\ f_2(x), & x \in U_2. \end{cases}$$

The agreement on overlaps guarantees that f is continuous.

Thus

$$C(-)$$

is a sheaf.

8.9 Counterexample: Bounded Continuous Functions

Define

$$\mathcal{F}(U) = \{f : U \to \mathbb{R} \mid f \text{ continuous and bounded}\}.$$

This is a presheaf.

Cover

$$\mathbb{R} = \bigcup_{n=1}^{\infty} (-n, n).$$

For each n,

$$f_n(x) = x$$

is bounded on $(-n, n)$.

These local sections agree on overlaps.

The glued function is

$$f(x) = x.$$

But f is not bounded on $\mathbb{R}$.

Hence the glued section does not belong to

$$\mathcal{F}(\mathbb{R}).$$

Therefore

$$\boxed{\mathcal{F} \text{ is not a sheaf.}}$$

8.10 Counterexample: Constant Functions

Define

$$\mathcal{F}(U) = \{\text{constant functions on } U\}.$$

Take

$$U = (-1, 0) \cup (0, 1).$$

Define

$$f_1 = 0$$

on $(-1, 0)$,

and

$$f_2 = 1$$

on $(0, 1)$.

There is no overlap.

Compatibility is automatic.

The glued function becomes

$$f = \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases}$$

which is not constant.

Hence constant functions form only a presheaf.

8.11 Quotient Presheaves

Suppose

$$\mathcal{G} \subseteq \mathcal{F}$$

is a sub-presheaf.

The quotient presheaf is

$$(\mathcal{F}/\mathcal{G})(U) = \mathcal{F}(U)/\mathcal{G}(U).$$

Example 8.11.1. Continuous functions modulo constants:

$$C(U)/\mathbb{R}.$$

A quotient presheaf may fail to satisfy the sheaf axioms.

This motivates sheafification.

8.12 Sheafification

Definition 8.12.1. For every presheaf $\mathcal{F}$ there exists a sheaf

$$\mathcal{F}^+$$

and a morphism

$$\mathcal{F} \to \mathcal{F}^+$$

which is universal among all morphisms from $\mathcal{F}$ to sheaves.

Intuitively, sheafification:

- keeps the local sections,
- identifies sections that agree locally,
- adds all missing glued sections.

8.13 Germs

Fix a point

$$x \in X.$$

Suppose

$$s \in \mathcal{F}(U), \qquad t \in \mathcal{F}(V),$$

where

$$x \in U \cap V.$$

Definition 8.13.1. The sections s and t are equivalent at x,

$$s \sim_x t,$$

if there exists a neighborhood

$$W \subseteq U \cap V$$

containing x such that

$$s|_W = t|_W.$$

The equivalence class

$$[s]_x$$

is called the *germ* of s at x.

8.14 Intuition Behind Germs

A germ remembers only what happens arbitrarily close to x.

Everything away from x is ignored.

$$\boxed{\text{Germ} = \text{local behavior near a point.}}$$

8.15 Stalks

Definition 8.15.1. The stalk of a sheaf $\mathcal{F}$ at x is

$$\mathcal{F}_x = \varinjlim_{x \in U} \mathcal{F}(U).$$

Equivalently,

$$\boxed{\mathcal{F}_x = \{\text{all germs at } x\}.}$$

8.16 Stalk as a Colimit

The neighborhoods of x

$$U_1 \supseteq U_2 \supseteq U_3 \supseteq \cdots$$

form a directed system.

$$\begin{tikzcd}
\mathcal{F}(U_1) \arrow[d] \arrow[drr] & & \\
\mathcal{F}(U_2) \arrow[d] \arrow[rr] & & \mathcal{F}_x \\
\mathcal{F}(U_3) \arrow[urr] & &
\end{tikzcd}$$

The colimit identifies sections that become equal on some smaller neighborhood.

8.17 Example of a Stalk

Let

$$\mathcal{F}(U) = C(U).$$

At $x = 0$,

$$f(x) = x$$

and

$$h(x) = \begin{cases} x, & |x| < 1, \\ 100, & |x| \geq 1 \end{cases}$$

define the same germ.

Indeed, on a sufficiently small neighborhood of 0,

$$f = h.$$

Hence

$$[f]_0 = [h]_0.$$

8.18 Conceptual Hierarchy

$$\begin{tikzcd}
\text{Topological Space} \arrow[d] \\
\text{Presheaf} \arrow[d] \\
\text{Sheaf} \arrow[d] \\
\text{Sections} \arrow[d] \\
\text{Germs} \arrow[d] \\
\text{Stalks} \arrow[d] \\
\text{Local Information}
\end{tikzcd}$$

8.19 Fundamental Principle

$$\boxed{\text{Presheaf} = \text{local objects with restrictions.}}$$

$$\boxed{\text{Sheaf} = \text{local objects that glue uniquely.}}$$

$$\boxed{\text{Germ} = \text{local behavior at one point.}}$$

$$\boxed{\text{Stalk} = \text{the set of all germs at that point.}}$$

$$\boxed{\mathcal{F}_x = \varinjlim_{x \in U} \mathcal{F}(U).}$$

The language of sheaves and stalks is one of the central foundations of modern algebraic geometry, complex geometry, differential geometry, cohomology theory, and the theory of schemes.

8.20 Additional Examples of Quotient Sheaves

Quotient sheaves arise whenever a sheaf contains a subsheaf and we identify sections differing by elements of that subsheaf.

Let

$$\mathcal{G} \subseteq \mathcal{F}$$

be a subsheaf.

The quotient sheaf is denoted

$$\mathcal{F}/\mathcal{G}.$$

For each open set U,

$$(\mathcal{F}/\mathcal{G})(U) = \mathcal{F}(U)/\mathcal{G}(U)$$

followed (if necessary) by sheafification.

Example 1: Continuous Functions Modulo Constants

Let

$$\underline{\mathbb{R}}$$

denote the constant sheaf.

Consider

$$\mathcal{C}/\underline{\mathbb{R}}.$$

For an open set U,

$$(\mathcal{C}/\underline{\mathbb{R}})(U) = C(U)/\mathbb{R}.$$

Two functions are identified whenever they differ by a constant:

$$f \sim g \quad \Longleftrightarrow \quad f - g = c.$$

For example,

$$x^2$$

and

$$x^2 + 5$$

represent the same section.

This quotient remembers only changes in the function and ignores vertical translations.

Example 2: Holomorphic Functions Modulo Constants

On a complex manifold let

$$\mathcal{O}$$

be the sheaf of holomorphic functions.

Then

$$\mathcal{O}/\underline{\mathbb{C}}$$

identifies holomorphic functions differing by constants.

Thus

$$z^2$$

and

$$z^2 + 7$$

represent the same class.

This quotient appears naturally when studying differentials because

$$d(f + c) = df.$$

Example 3: Functions Modulo Vanishing Functions

Fix a point

$$p \in X.$$

Let

$$\mathfrak{m}_p(U) = \{f \in \mathcal{O}(U) \mid f(p) = 0\}.$$

Then

$$\mathcal{O}_p/\mathfrak{m}_p$$

is called the residue field at p.
For complex varieties:

$$\mathcal{O}_p/\mathfrak{m}_p \cong \mathbb{C}.$$

This quotient remembers only the value of a function at the point p.
Indeed,

$$f \equiv g \pmod{\mathfrak{m}_p}$$

iff

$$f(p) = g(p).$$

Example 4: First-Order Infinitesimal Quotient

Let

$$\mathfrak{m}_p$$

be the maximal ideal of functions vanishing at p.
Consider

$$\mathfrak{m}_p/\mathfrak{m}_p^2.$$

This quotient forgets all quadratic and higher-order terms.
For

$$f(x) = 3x + 5x^2 + x^3,$$

the class of f in

$$\mathfrak{m}_p/\mathfrak{m}_p^2$$

is represented simply by

$$3x.$$

This space is dual to the tangent space:

$$T_p^*X \cong \mathfrak{m}_p/\mathfrak{m}_p^2.$$

It is one of the most important quotients in algebraic geometry.

Example 5: Differential Forms Modulo Exact Forms

Let

$$\Omega^1$$

be the sheaf of differential 1-forms.
Let

$$d\mathcal{O}$$

denote the image of

$$d : \mathcal{O} \to \Omega^1.$$

Then

$$\Omega^1/d\mathcal{O}$$

identifies forms differing by an exact form.
For example,

$$(x\,dx)$$

and

$$(x\,dx + d(x^2))$$

represent the same class.
Such quotients lead naturally to de Rham cohomology.

Example 6: Rational Functions Modulo Regular Functions

Let

$$\mathcal{K}$$

be the sheaf of rational functions on a variety.

Since

$$\mathcal{O} \subseteq \mathcal{K},$$

we may form

$$\mathcal{K}/\mathcal{O}.$$

A section records only the poles of a rational function.

For instance,

$$\frac{1}{x}$$

defines a nontrivial class because it is not regular at $x = 0$.

This quotient plays a central role in divisor theory.

Example 7: The Exponential Sequence Quotient

Consider the exponential map

$$\exp(2\pi i\cdot) : \mathcal{O} \to \mathcal{O}^\times.$$

Its kernel is

$$\underline{\mathbb{Z}}.$$

Thus there is an exact sequence

$$0 \to \underline{\mathbb{Z}} \to \mathcal{O} \to \mathcal{O}^\times \to 0.$$

Here

$$\mathcal{O}^\times \cong \mathcal{O}/\underline{\mathbb{Z}}$$

(after interpreting the quotient appropriately through the exact sequence).

This leads directly to line bundles and Chern classes.

8.21 Geometric Interpretation of Important Quotients

Quotient	What Is Forgotten	What Remains
$C/\mathbb{R}$	Constant shifts	Relative shape
$\mathcal{O}/\mathbb{C}$	Constant terms	Holomorphic variation
$\mathcal{O}_p/\mathfrak{m}_p$	Everything except value	Function value at p
$\mathfrak{m}_p/\mathfrak{m}_p^2$	Quadratic and higher terms	Linear information
$\Omega^1/d\mathcal{O}$	Exact forms	Cohomological information
$\mathcal{K}/\mathcal{O}$	Regular part	Pole structure